

Continuous r.v.

①. $X \sim$ exponential r.v.

$$\text{var}(X) = 25$$

a. PDF of X

b. $E(X^2) = ?$

c. $P(X > 5) = ?$

$$\text{a. } X \sim \exp(\lambda) \Rightarrow \text{var}(X) = \frac{1}{\lambda^2} = 25 \Rightarrow \lambda^2 = \frac{1}{25} \mid \lambda > 0 \Rightarrow \boxed{\lambda = \frac{1}{5}}$$

$$\Rightarrow f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ \text{zero,} & \text{otherwise} \end{cases}$$

$$f(x) = \begin{cases} \frac{1}{5} e^{-\frac{1}{5}x}, & x \geq 0 \\ \text{zero,} & \text{otherwise} \end{cases} \quad - \text{ PDF } f(x)$$

$$\text{b. } E(X) = \int_{-\infty}^{+\infty} x \cdot f(x) dx \Rightarrow E(X^2) = \int_{-\infty}^{+\infty} x^2 \cdot f(x) dx$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^0 x^2 \cdot 0 dx + \int_0^{\infty} x^2 \cdot \frac{1}{5} e^{-\frac{1}{5}x} dx = \\ &= \frac{1}{5} \lim_{t \rightarrow \infty} \int_0^t x^2 \cdot e^{-\frac{1}{5}x} dx \Rightarrow \end{aligned}$$

$$f(x) = x^2 \Rightarrow f'(x) = 2x$$

$$g'(x) = e^{-\frac{1}{5}x} \Rightarrow g(x) = -5e^{-\frac{1}{5}x} \quad | \Rightarrow$$

$$\Rightarrow \int_0^t x^2 \cdot e^{-\frac{1}{5}x} dx = -5x^2 \cdot e^{-\frac{1}{5}x} \Big|_0^t + 10 \int_0^t x \cdot e^{-\frac{1}{5}x} dx =$$

$$= -5t^2 \cdot e^{-\frac{1}{5}t} + 10 \int_0^t x \cdot e^{-\frac{1}{5}x} dx$$

$$\Rightarrow \underline{E(X^3)} = \frac{1}{5} \lim_{t \rightarrow \infty} \left(-5t^2 \cdot e^{-\frac{1}{5}t} + 10 \cdot 5 \int_0^t x \cdot \frac{1}{5} \cdot e^{-\frac{1}{5}x} dx \right)$$

$$= \frac{1}{5} \cdot 10 \cdot 5 \cdot \int_0^{\infty} x \cdot f(x) dx = 10 F(x) =$$

$$= 10 \cdot \frac{1}{\frac{1}{5}} = 10 \cdot 5 = \boxed{50}$$

$$c. P(X > 5) = 1 - P(X \leq 5) = 1 - \int_{-\infty}^5 f_x(x) dx \quad \text{OR} \quad 1 - F_x(5)$$

$$X \sim \exp(\lambda = \frac{1}{5})$$

$$CDF \Rightarrow F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x > 0 \\ \text{zero,} & \text{otherwise} \end{cases}$$

$$F_X(x) = \begin{cases} 1 - e^{-\frac{1}{5}x}, & x > 0 \\ \text{zero,} & \text{otherwise} \end{cases}$$

$$\Rightarrow \underline{P(X > 5)} = 1 - F_X(5) = 1 - 1 + e^{-\frac{1}{5} \cdot 5} = \boxed{e^{-1} = \frac{1}{e}}$$

② X-r.v. with PDF: $f_X(x) = \begin{cases} \frac{\lambda^n \cdot x^{n-1} \cdot e^{-\lambda x}}{(n-1)!}, & x \geq 0 \\ \text{zero,} & \text{otherwise} \end{cases}$ (Erwartungswert) $n \in \mathbb{N}$

For $\lambda = \frac{1}{3}$, det:

a. CDF for $n=3$

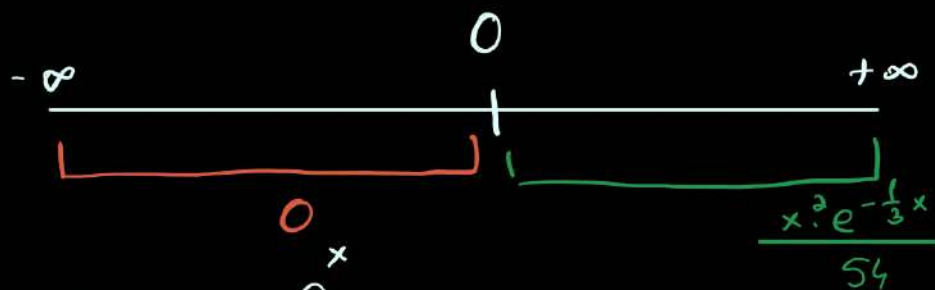
b. $\text{Var}(X) = ?$, $n=3$

c. $n = ?$ such that $E(X) = 15 \rightarrow \text{Homework}$

a. CDF: $F_X(x) = \int_{-\infty}^x f_X(x) dx$

$\left. \begin{matrix} n=3 \\ \lambda = \frac{1}{3} \end{matrix} \right\} \Rightarrow f_X(x) = \begin{cases} \frac{(\frac{1}{3})^3 \cdot x^2 \cdot e^{-\frac{1}{3}x}}{2}, & x \geq 0 \\ \text{zero} & , \text{ otherwise} \end{cases}$

$f_X(x) = \begin{cases} \frac{x^2 e^{-\frac{1}{3}x}}{54}, & x \geq 0 \\ \text{zero} & , \text{ otherwise} \end{cases}$



• $x < 0$ $\Rightarrow F_X(x) = \int_{-\infty}^x 0 dx = 0$

• $x > 0$ $\Rightarrow F_X(x) = \int_{-\infty}^0 \underline{0} dt + \int_0^x \frac{t^2 \cdot e^{-\frac{1}{3}t}}{54} dt$
 $= \frac{1}{54} \int_0^x t^2 \cdot e^{-\frac{1}{3}t} dt$

$$f(x) = t^2 \Rightarrow f'(x) = 2t$$

$$g'(x) = e^{-\frac{1}{3}t} \Rightarrow g(x) = -3e^{-\frac{1}{3}t}$$

$$f(x) = t \Rightarrow f'(x) = 1$$

$$g'(x) = e^{-\frac{1}{3}t} \Rightarrow g(x) = -3e^{-\frac{1}{3}t}$$

$$\Rightarrow F_X(x) = \frac{1}{54} \left(-3t^2 \cdot e^{-\frac{1}{3}t} \Big|_0^x + 6 \int_0^x t \cdot e^{-\frac{1}{3}t} dt \right) =$$

$$= \frac{1}{54} \left(-3x^2 \cdot e^{-\frac{1}{3}x} + 6 \left(-3t \cdot e^{-\frac{1}{3}t} \Big|_0^x + 3 \int_0^x e^{-\frac{1}{3}t} dt \right) \right)$$

$$= \frac{1}{54} \left(-3x^2 \cdot e^{-\frac{1}{3}x} + 6 \left(-3x \cdot e^{-\frac{1}{3}x} + 3 \int_0^x e^{-\frac{1}{3}t} dt \right) \right)$$

$$= \frac{1}{54} \left(-3x^2 \cdot e^{-\frac{1}{3}x} - 18x \cdot e^{-\frac{1}{3}x} - 54e^{-\frac{1}{3}t} \Big|_0^x \right) =$$

$$= \frac{1}{54} \left(3x^2 \cdot e^{-\frac{1}{3}x} - 18x e^{-\frac{1}{3}x} - 54e^{-\frac{1}{3}x} + 54 \right)$$

$$\Rightarrow F_X(x) = \begin{cases} \frac{1}{54} \left(3x^2 \cdot e^{-\frac{1}{3}x} - 18x e^{-\frac{1}{3}x} - 54e^{-\frac{1}{3}x} + 54 \right), & x \geq 0 \\ \text{zero, otherwise} \end{cases}$$

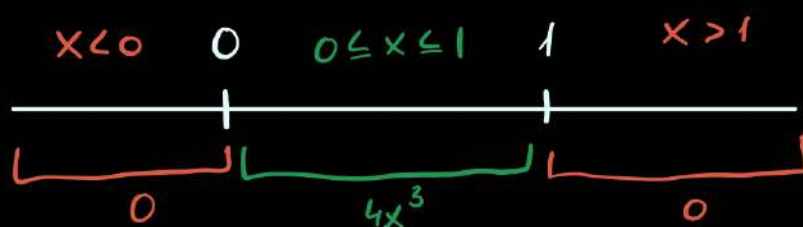
- ③ X - r.v with PDF: $f(x) = \begin{cases} c \cdot x^3, & x \in [0, 1] \\ \text{zero, otherwise} \end{cases}$
- $c = ?$
 - CDF = ?
 - $P(X = 0.5) = ?$ $P(X > \frac{1}{2}) = ?$ $P_{x > \frac{1}{3}}(X \leq \frac{2}{3}) = ?$
 - $E(X) = ?$ $\text{Var}(X) = ?$

a. $\int_{-\infty}^{\infty} \underline{f(x)} dx = 1$ and $\underline{f(x) \geq 0} \Rightarrow c \geq 0$

$$\int_{-\infty}^0 \underline{0} dx + \int_0^1 c \cdot x^3 dx + \int_1^{\infty} \underline{0} dx = c \int_0^1 x^3 dx = c \cdot \frac{x^4}{4} \Big|_0^1 = c \cdot \frac{1}{4} = 1 \Rightarrow \boxed{c=4}$$

$$\Rightarrow f(x) = \begin{cases} 4x^3, & x \in [0, 1] \\ \text{zero, otherwise} \end{cases}$$

b. $F_X(x) = \int_{-\infty}^x f(t) dt$



• $x < 0$ $\Rightarrow F_X(x) = \int_{-\infty}^0 0 dt = 0$

• $x \in [0, 1]$ $\Rightarrow F_X(x) = \int_{-\infty}^0 \underline{0} dt + \int_0^x 4t^3 dt = 4 \int_0^x t^3 dt = 4 \cdot \frac{t^4}{4} \Big|_0^x = x^4$

• $x > 1$ $\Rightarrow F_X(x) = \int_{-\infty}^0 \underline{0} dt + \int_0^1 4t^3 dt + \int_1^x \underline{0} dt = 4 \int_0^1 t^3 dt = 4 \cdot \frac{t^4}{4} \Big|_0^1 = 1$

CDF: $F_X = \begin{cases} 0, & x < 0 \\ x^4, & x \in [0, 1] \\ 1, & x > 1 \end{cases}$

c. $P_B(A) = \frac{P(A \cap B)}{P(B)}$

$$\begin{aligned}
 P_{x > \frac{1}{3}} \left(X \leq \frac{2}{3} \right) &= \frac{P\left(X \leq \frac{2}{3} \cap X > \frac{1}{3}\right)}{P\left(X > \frac{1}{3}\right)} = \frac{P\left(\frac{1}{3} < X \leq \frac{2}{3}\right)}{P\left(X > \frac{1}{3}\right)} = \\
 &= \frac{F\left(\frac{2}{3}\right) - F\left(\frac{1}{3}\right)}{1 - P\left(X \leq \frac{1}{3}\right)} = \frac{F\left(\frac{2}{3}\right) - F\left(\frac{1}{3}\right)}{1 - F\left(\frac{1}{3}\right)} = \\
 &= \frac{\left(\frac{2}{3}\right)^4 - \left(\frac{1}{3}\right)^4}{1 - \left(\frac{1}{3}\right)^4}
 \end{aligned}$$

$$P(X = 0,5) = 0$$

$$P\left(X > \frac{1}{2}\right) = 1 - P\left(X \leq \frac{1}{2}\right) = 1 - F\left(\frac{1}{2}\right) = 1 - \left(\frac{1}{2}\right)^4$$

$$d. E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_{-\infty}^0 x \cdot 0 dx + \int_0^1 x \cdot 4x^3 dx + \int_1^{+\infty} x \cdot 0 dx$$

$$E(X) = 4 \cdot \frac{x^5}{5} \Big|_0^1 = \frac{4}{5}$$

$$\underline{Var(X) = E(X^2) - [E(X)]^2 \rightarrow \text{Homework}}$$